Essential Summary

Generalized Projective Synchronization of Chaotic Systems via Modified Active Control

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1 Background and Motivation

Since Pecora and Carroll's 1990 demonstration of chaos synchronization, the field has grown into a rich hierarchy of synchronization types — complete, phase, lag, generalized, and projective synchronization. **Generalized projective synchronization (GPS)** stands out because it relaxes two constraints of earlier projective synchronization: the systems no longer need to be partially linear, and the drive and response systems need not share the same structure. The defining feature of GPS is a projection matrix αI: the response state converges to a *scaled* replica of the drive state.

$$\lim_{t\to\infty}\|\mathbf{e}(t)\| = \lim_{t\to\infty}\|\mathbf{x}_m(t) - \alpha\mathbf{x}_s(t)\| = 0 \tag{1}$$

The scaling factor α allows arbitrary "compression" ($|\alpha| < 1$) or "stretching" ($|\alpha| > 1$) of the drive signal, in-phase ($\alpha > 0$) or anti-phase ($\alpha < 0$). This tunable property makes GPS attractive for digital secure communications.

The classical **active control** approach builds the controller matrix M by requiring all eigenvalues of the closed-loop matrix $E - M$ to have negative real parts — the **Routh–Hurwitz criterion**. For high-dimensional systems, computing these eigenvalues is expensive and error-prone. This paper proposes a **modified active control** method that replaces eigenvalue analysis with a **special matrix construction** validated directly by Lyapunov stability theory, thereby eliminating the Routh–Hurwitz dependence entirely.

2 Problem Formulation

Consider the drive system

$$\dot{\mathbf{x}}_m = A\mathbf{x}_m + Bf(\mathbf{x}_m), \tag{2}$$

and the response system

$$\dot{\mathbf{x}}_s = C\mathbf{x}_s + Dg(\mathbf{x}_s) + \mathbf{u}, \tag{3}$$

where A, B, C, D are constant matrices, f, g are nonlinear maps, and $\mathbf{u}$ is the controller to be designed. Define the error $\mathbf{e} = \mathbf{x}_m - \alpha\mathbf{x}_s$. Subtracting Eq. (3) from Eq. (2) yields the error system

$$\dot{\mathbf{e}} = E\mathbf{e} + Kh(\mathbf{x}_m, \mathbf{x}_s, \alpha) - \alpha\mathbf{u}, \tag{4}$$

with constant matrices E, K and nonlinear residual h. GPS is achieved if a controller $\mathbf{u}$ renders the origin of Eq. (4) asymptotically stable.

3 Modified Active Control

3.1 Controller Structure

The controller is decomposed as $\mathbf{u} = \mathbf{u}_a + \mathbf{u}_b$:

$$\boxed{\mathbf{u}_a = \alpha^{-1} M\mathbf{e}, \qquad \mathbf{u}_b = \alpha^{-1} Kh(\mathbf{x}_m, \mathbf{x}_s, \alpha)} \tag{5}$$

The component $\mathbf{u}_b$ cancels the nonlinear residual Kh, while $\mathbf{u}_a$ shapes the linear part. Substituting Eq. (5) into Eq. (4) collapses the error dynamics to the linear system

$$\dot{\mathbf{e}} = (E - M)\,\mathbf{e}. \tag{6}$$

3.2 Special Matrix Construction (Key Idea)

The classical method requires $\lambda(E - M) \subset \mathbb{C}^-$ (Routh–Hurwitz). The modified method instead chooses M so that the closed-loop matrix $N = E - M$ satisfies **only**:

- $n_{ij} = -n_{ji}$ for $i \neq j$ (skew-symmetric off-diagonal blocks),
- $n_{ii} \leq 0$, not all zero (non-positive diagonal),
- there exist positive weights $k_i > 0$.

Why this works: choose the Lyapunov function

$$V = \sum_{i=1}^{n} \frac{e_i^2}{k_i}, \qquad \dot{V} = 2\sum_{i=1}^{n} \frac{n_{ii}}{k_i} e_i^2 \;\leq\; 0, \tag{7}$$

so asymptotic stability follows immediately without any eigenvalue computation. For high-dimensional systems this is dramatically simpler than Routh–Hurwitz.

4 Example 1: Self-Structure Synchronization (Energy Resource System)

The energy demand–supply system (describing the coupled relationship between Jiangsu's energy demand and Western energy supply) is used as both drive and response:

$$\dot{x}_m = a_1 x_m[1 - x_m/M] - a_2(y_m + z_m), \qquad \dot{y}_m = -b_1 y_m - b_2 z_m + b_3 x_m[N - (x_m - z_m)], \qquad \dot{z}_m = c_1 z_m(c_2 x_m - c_3), \tag{8}$$

with $a_1 = 0.09,\ a_2 = 0.15,\ b_1 = 0.06,\ b_2 = 0.082,\ b_3 = 0.07,\ c_1 = 0.2,\ c_2 = 0.5,\ c_3 = 0.4,\ M = 1.8,\ N = 1$. The resulting chaotic attractor is shown in Figure 1.

For $\alpha = -2$ (anti-phase compression) and $\alpha = 3$ (in-phase compression), the controller Eq. (5) with $k_1 = 1,\ k_2 = 5,\ k_3 = 10$ drives the error to zero, as shown in Figures 2–3.

5 Example 2: Heterostructure Synchronization (Energy System → NSG)

The energy resource system drives a structurally different **Nuclear Spin Generator (NSG)** system:

$$\dot{x}_s = -a_3 x_s + y_s + u_1, \qquad \dot{y}_s = -x_s - a_3 y_s(1 - b_4 z_s) + u_2, \qquad \dot{z}_s = c_4 a_3(1 - z_s) - a_3 b_4 y_s^2 + u_3, \tag{9}$$

demonstrating that the modified active control works for **heterostructure** GPS — the drive and response systems share neither structure nor parameter set.

The scaling law is complete: $-1 < \alpha < 0$ gives anti-phase stretching, $0 < \alpha < 1$ in-phase stretching, and $\alpha = \pm 1$ recovers complete and anti-phase synchronization as special cases.

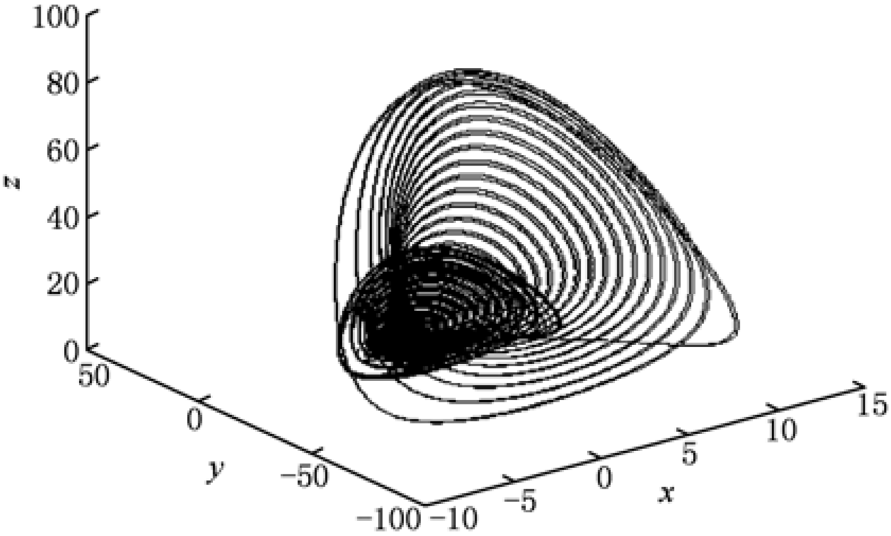

Figure 1: **Figure 1.** Three-dimensional chaotic phase portrait of the energy resource system Eq. (8).

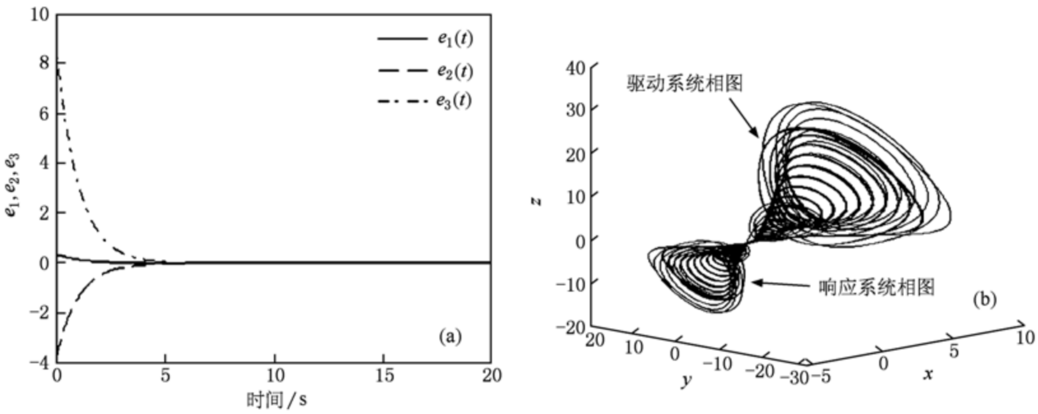

Figure 2: **Figure 2.** (a) Synchronization error curves for $\alpha = -2$; (b) three-dimensional GPS phase portraits of drive and response systems for $\alpha = -2$. The response phase portrait is an anti-phase compressed replica of the drive.

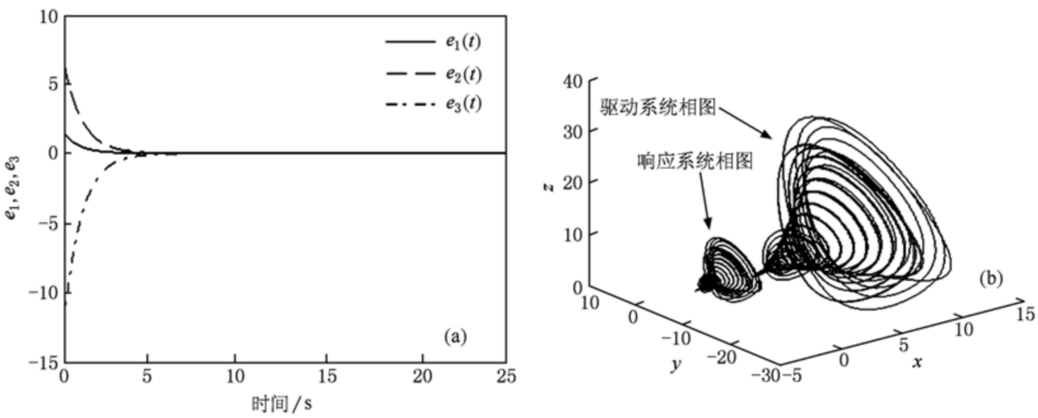

Figure 3: **Figure 3.** (a) Synchronization error curves for $\alpha = 3$; (b) three-dimensional GPS phase portraits for $\alpha = 3$, showing in-phase compression.

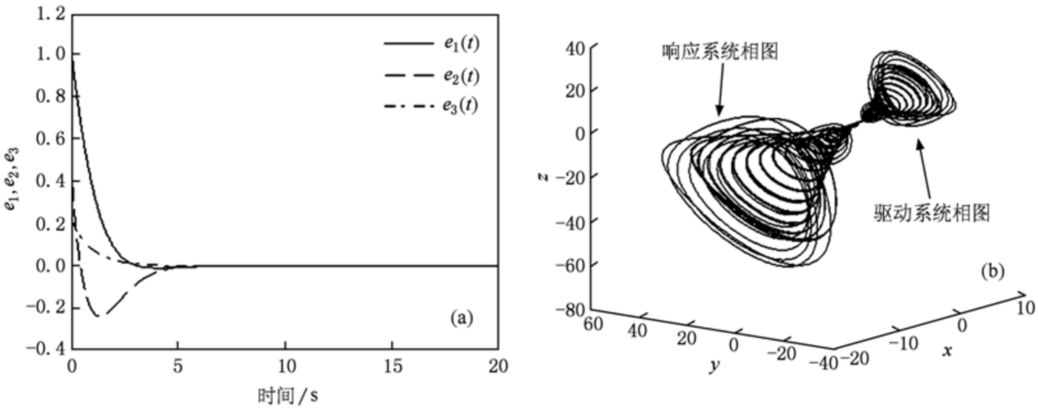

Figure 4: **Figure 4.** (a) Heterostructure synchronization error for $\alpha = -0.5$ (anti-phase stretch); (b) three-dimensional GPS phase portraits for $\alpha = -0.5$.

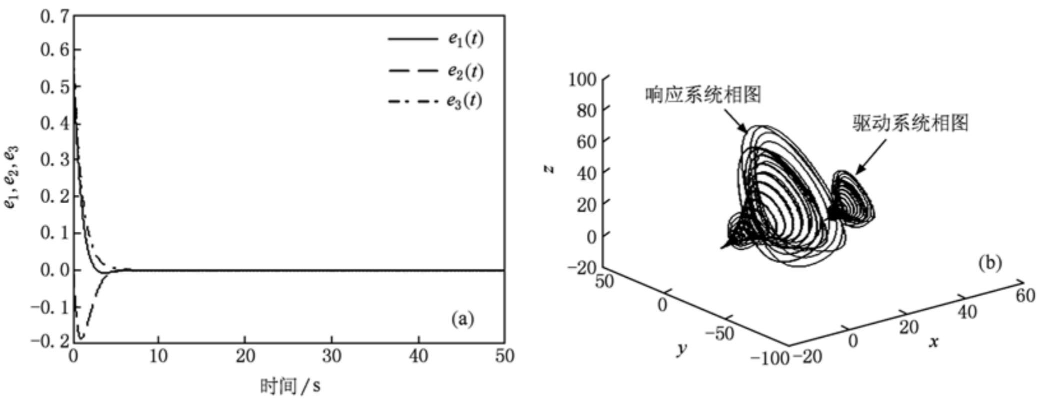

Figure 5: **Figure 5.** (a) Heterostructure synchronization error for $\alpha = 0.4$ (in-phase stretch); (b) three-dimensional GPS phase portraits for $\alpha = 0.4$.

6 Comparison with Tracking Control

Figure 6 shows the synchronization error produced by the classical tracking-control approach for the same drive–response pair ($\alpha = -0.5$). Compared with Figure 4(a), the modified active control converges **noticeably faster**, confirming its higher synchronization efficiency.

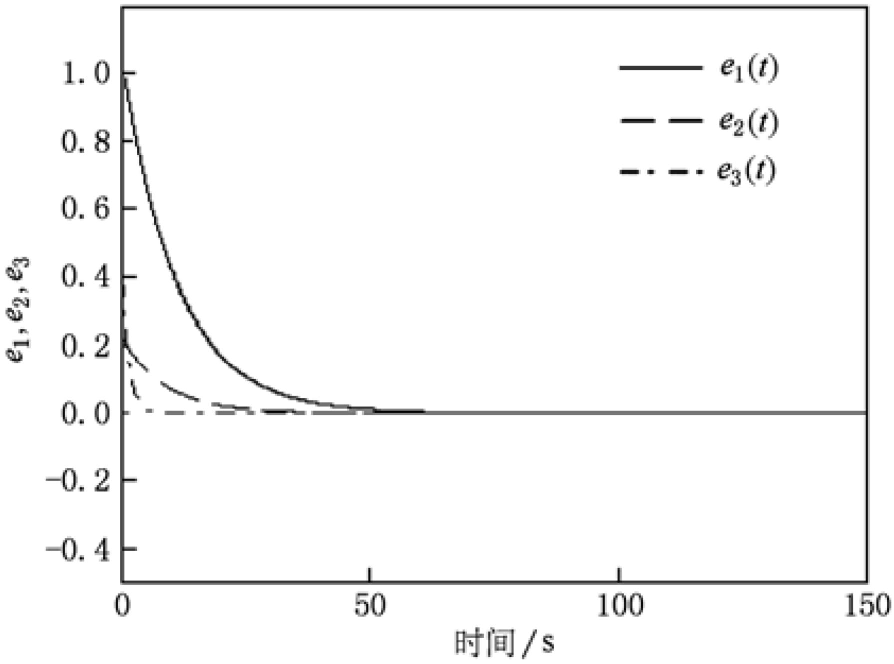

Figure 6: **Figure 6.** Synchronization error curves under the tracking-control method. Faster convergence in Figure 4 demonstrates the efficiency advantage of modified active control.

7 Key Findings

1. **Routh–Hurwitz independence (core contribution):** By constructing a special matrix M so that the closed-loop matrix $N = E - M$ is skew-symmetric off-diagonal with non-positive diagonal, global GPS is guaranteed through the Lyapunov function Eq. (7) *without* computing eigenvalues. This dramatically simplifies high-dimensional synchronization design.

2. **Universal applicability:** The method is validated for both self-structure GPS (energy system, $\alpha = -2, 3$) and heterostructure GPS (energy system $\rightarrow$ NSG system, $\alpha = -0.5, 0.4$), and works for low- and high-dimensional systems alike.

3. **Flexible scaling:** Tuning α realizes arbitrary in-phase/anti-phase compression or stretching of the drive signal — a property valuable for secure communications.

4. **Higher efficiency than tracking control:** Error convergence under modified active control is faster than under tracking control, while retaining the robustness and simplicity of classical active control.

5. **Theoretical–numerical agreement:** All numerical simulations confirm the theoretical analysis, establishing the method's reliability.

8 Significance

This paper represents the author's early work in nonlinear dynamics, centered on **chaos synchronization and control**. The two technical ingredients — Lyapunov stability analysis and structured matrix construction — later informed the author's development of analytical perturbation methods (the GHFP/MGHFP series). The paper also illustrates a general principle: for synchronization control, a well-chosen Lyapunov function can replace expensive algebraic criteria such as Routh–Hurwitz, making controller design simultaneously simpler and more transparent.

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